RESEARCH ARTICLE

Previous Antiplatelet Therapy and Outcomes of Acute Ischemic Stroke With Large Vessel Occlusion Treated With Direct Endovascular Therapy

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Abstract

Background and Objectives

Previous antiplatelet therapy (APT) may influence outcomes in patients undergoing direct endovascular therapy (EVT) for anterior circulation large vessel occlusion (LVO) stroke, but evidence on clinical benefits and safety is limited and inconsistent. We aimed to evaluate the effects of previous APT on the outcomes of direct EVT.

Methods

We conducted a retrospective analysis of consecutive patients treated at 20 high-volume stroke centers across 9 European countries and Israel (EVA-TRISP registry, 2015–2023). We included adults with anterior circulation LVO strokes, without previous use of IV thrombolysis (IVT) or anticoagulants. We compared outcomes of direct EVT patients stratified by previous APT regimen, using propensity score matching based on medical history and multilevel models to address confounders. The primary outcome was the 90-day modified Rankin Scale (mRS) score. The secondary efficacy outcomes were 90-day independence and the rate of successful reperfusion. The secondary safety outcomes were the rate of symptomatic intracranial hemorrhage (sICH) and mortality.

Results

Among 12,950 patients, 2,611 met the criteria and 1,308 were matched: 480 without previous APT and 828 with any previous APT, among whom 764 were on single APT and 64 on dual APT. The overall mean age was 75.8 ± 11.4 years, and 49.4% were female; the mean NIH Stroke Scale score was 13.1 ± 7.2 points, the successful reperfusion rate was 74.7%, and the median mRS score was 3 (interquartile range 2–4), with 38.8% of patients independent at 90 days. When compared with no previous APT, any previous APT was associated with a shift toward lower mRS scores at 90 days (odds ratio [OR] 1.30, CI 1.04–1.61, $p = 0.018$). Independence at 90 days was associated with any previous APT (OR 1.62, CI 1.22–2.16, $p = 0.001$). Any previous APT was not associated with successful reperfusion (OR 0.96, CI 0.69–1.35, $p = 0.821$), sICH (OR 1.06, CI 0.47–2.39, $p = 0.880$), or mortality (OR 0.89, CI 0.66–1.21, $p = 0.821$). There was no significant interaction between any previous APT and proximal/distal occlusion location ($p = 0.213$) or time-to-groin earlier/later than 6 hours ($p = 0.743$).

Discussion

In patients with anterior circulation LVO stroke treated with direct EVT and no previous anticoagulation, pretreatment with any APT was independently linked to better 90-day functional outcomes and higher independence, without increasing sICH risk.

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Class of Evidence
Criteria for rating therapeutic and diagnostic studies
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Supplementary Material

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Glossary

APT = antiplatelet therapy; **DAPT** = dual APT; **EVT** = endovascular therapy; **ICA** = internal carotid artery; **IQR** = interquartile range; **IVT** = IV thrombolysis; **LVO** = large vessel occlusion; **mRS** = modified Rankin Scale; **mTICI** = modified Thrombolysis in Cerebral Infarction; **OR** = odds ratio; **SAPT** = single APT; **sICH** = symptomatic intracranial hemorrhage.

Classification of Evidence

This study provides Class III evidence that, in patients with anterior circulation LVO stroke treated with direct EVT, previous APT is associated with better 90-day functional outcomes compared with no previous APT.

Introduction

There is conflicting evidence that previous antiplatelet therapy (APT) influences the outcome of patients with acute ischemic stroke who are treated with endovascular therapy (EVT) with or without IV thrombolysis (IVT). While most retrospective studies reported no effects on the 90-day modified Rankin Scale (mRS) score, the successful reperfusion rate, or the risk of bleeding,[1-6] others reported functional benefits without an increased risk of symptomatic intracranial hemorrhage (sICH).[7-11] The only randomized controlled trial that evaluated the effects of a loading dose of APT and/or prophylactic anticoagulation given intravenously in the acute setting before EVT (with IVT in 75% of cases) was prematurely stopped because of safety concerns. Indeed, acute APT was associated with a higher risk of sICH, without apparent influence on the 90-day outcome.[12]

Dedicated studies on the effect of previous APT on the outcomes of direct EVT—when IVT is not administered—are scarce. Direct EVT is performed for acute ischemic stroke of accessible (large) vessel occlusion, when IVT is contraindicated, usually because of a late stroke presentation (>4.5 hours) that limits the use of fibrinolysis or because the risk of bleeding is too high. A retrospective study that included a subgroup analysis of patients on previous APT treated with direct EVT reported nonsignificant findings on functional outcome and sICH risk.[1] A single randomized controlled trial that evaluated the effects of the IV antiplatelet agent tirofiban on the outcomes of direct EVT (without the use of previous IVT) showed nonsignificant difference in disability and sICH risk compared with placebo (no APT).[13]

We aimed to evaluate the influence of previous APT on the outcomes of direct EVT. The primary research question was whether, in patients with anterior circulation large vessel occlusion (LVO) stroke treated with direct EVT, previous APT was associated with better 90-day functional outcomes compared with no previous APT.

Methods

Study Design and Setting

In this observational study, we conducted a retrospective analysis of prospectively collected data between 2015 and 2023 as part of the EVA-TRISP database. The EVA-TRISP registry is a large, multicenter collaboration designed to generate evidence on the safety and efficacy of IVT and EVT for acute ischemic stroke. The collaboration includes 20 high-volume stroke centers across 9 European countries (Switzerland, the Netherlands, Germany, Sweden, Finland, Italy, France, Serbia) and Israel. All centers systematically collect consecutive data on patients treated with IVT and/or EVT, ensuring broad representativeness and minimizing selection bias. The data set contains more than 100 predefined variables, including demographics, prestroke disability, previous medications, vascular risk factors, laboratory and imaging findings, stroke etiology and clinical severity, procedural characteristics, and short-term and long-term outcomes. Outcome assessment is harmonized across centers, and all data sets are fully anonymized.[14]

Participants

We included in the analysis patients older than 18 years without severe disability (baseline mRS score ≤3 regardless of the cause), who had ischemic stroke due to LVO of the anterior circulation (internal carotid artery [ICA], middle cerebral artery [M1, M2], anterior cerebral artery [A1]) and were treated with direct EVT. Patients with extracranial carotid stenosis/occlusion were included only in the presence of an intracranial LVO. We excluded patients treated with bridging (IVT + EVT). In addition, we excluded patients diagnosed with posterior circulation strokes, namely with occlusion of the vertebral, basilar, and posterior cerebral arteries, as well as patients with isolated occlusion of the extracerebral carotids, distal (M3, M4) segment of the middle cerebral, or distal anterior (A2) arteries. Finally, patients who were on anticoagulants at the time of the stroke event and patients with missing information about the following variables were also excluded: baseline mRS score, 90-day mRS score, pharmacotherapy (specifically antiplatelets, anticoagulants, antihypertensives, and statins), age, sex, and medical history (previous stroke, atrial fibrillation, diabetes, hypertension, hypercholesterolemia). Previous single APT (SAPT) was defined as ongoing treatment with either aspirin, clopidogrel, prasugrel, ticagrelor, or dipyridamole at the time of stroke onset, and previous dual APT (DAPT) was defined as any combination of these drugs. Any previous APT was defined as either previous SAPT or DAPT. Stroke etiology was based on the Trial of Org 10172 in Acute Stroke Treatment classification.[15]

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Outcome Measures

The primary outcome was the 90-day mRS score after stroke (shift analysis). The secondary efficacy outcomes were patients' independence (90-day mRS score 0–2) and the rate of successful reperfusion (modified Thrombolysis in Cerebral Infarction [mTICI] score ≥2B). The secondary safety outcomes were mortality at 90 days after stroke and the rate of in-hospital sICH according to the criteria from the European Cooperative Acute Stroke Study II.[16] All outcome measures were ascertained by qualified clinicians across all centers. Information on successful reperfusion using the mTICI score was provided by the interventional radiologist performing EVT at the end of the procedure. The 90-day outcomes were assessed by certified clinicians or nurses who were unaware of previous APT status.

Statistical Analyses

Quantitative variables were expressed as mean and SD, if normality of distribution was confirmed with the Shapiro-Wilk test. Ordinal variables were expressed as median and interquartile range (IQR). Categorical variables were expressed as counts and percentages. Patients were matched using nearest-neighbor propensity scores with replacement, meaning that the same control could be matched to more than 1 treated patient if it provided the closest match, to reduce bias due to unbalanced baseline characteristics. Propensity scores were calculated based on logistic regression to model the probability of having any previous APT, based on the following variables: age, sex, diabetes, hypertension, atrial fibrillation, hypercholesterolemia, previous stroke, therapy with statins, and antihypertensive agents. For adequate matching, we used a width of 0.2 times the SD of the logit of the propensity score, which reduces the overall sample size but ensures higher quality matches.[17,18] To estimate the contribution of previous APT and account for residual bias within the matched cohort, we implemented a cumulative link mixed-effects model on the primary outcome (with higher odds ratios [ORs] indicating a shift toward better 90-day mRS scores) and mixed-effects logistic regression on secondary outcomes (with higher OR indicating higher probability of the outcome of interest). We included the following variables as covariates in the regression models: age, sex, APT status, time-to-groin, stroke etiology (cardioembolism, large artery atherosclerosis, other causes), center, previous mRS score, and admission NIHSS score. Because the EVA-TRISP data set includes multiple centers across Europe, we expected possible heterogeneity in EVT protocols, device usage, post-procedural care, or all the above, and we, therefore, included the variable center as a random effect to mitigate center bias toward the primary outcome.

To assess the robustness of the results, we performed sensitivity analyses on the primary outcome by varying the following conditions: with and without adjusting for center[19] and with and without including the reasons for the exclusion from IVT as covariates, because both exclusion by "imaging criteria associated with increased risk of hemorrhage" and "exclusion by clinician-assessed bleeding risk" could influence the primary outcome. Finally, we performed a subgroup analysis with an interaction term based on the APT regimen (previous SAPT and DAPT vs no previous APT), occlusion location (proximal [ICA and M1] vs distal [M2]), and time-to-groin (early <6 hours vs late ≥ 6–24 hours). Statistical analyses were performed with RStudio version 2022.12.0.353 and the R packages "MatchIt," "MASS," "ordinal," and "lme4."

Standard Protocol Approvals, Registrations, and Patient Consents

Each center has received necessary official approval from their respective local authorities and/or ethical committees according to national and local rules. This article conforms to the Strengthening the Reporting of Observational Studies in Epidemiology reporting guideline and checklist. The ethics committee of the Canton of Ticino approved this study (registration number: Req-2024-00079).

Data Availability

Datasets generated or analyzed within this study are available from the corresponding author on reasonable request. In each such case, compliance of data sharing with individual processes of patient consenting in participating centers is required. Final decision on data sharing will be made by consensus of the EVA-TRISP collaborators.

Results

Patient Characteristics

Of 12,950 patients included in the EVA-TRISP database, 2,611 patients satisfied the inclusion and exclusion criteria for the study.

A total of 1,308 patients were matched by propensity scores: 36.7% (480/1,308) were not on previous APT, 63.3% (828/1,308) were on any previous APT, 58.4% (764/1,308) were on previous SAPT, and 4.9% (64/1,308) were on previous DAPT (Figure 1). Patients on previous SAPT took aspirin in most cases (671/764, 87.8%), followed by clopidogrel (92/764, 12.0%), and only 1 patient was on ticagrelor (1/764). Patients on previous DAPT took a combination of aspirin and another antiplatelet agent: clopidogrel in 71.9% (46/64), ticagrelor in 14.1% (9/64), dipyridamole in 9.3% (6/64), and prasugrel in 4.7% (3/64) of cases. The overall mean age was 75.8 ± 11.4 years, 49.4% were female, and the mean NIH Stroke Scale score was 13.1 ± 7.2 points. Table 1 lists baseline characteristics of patients included in the study after matching. We achieved negligible imbalance (standardized mean difference ≤0.1 for each variable) and good overlap of the propensity scores at visual inspection. The standardized mean differences before and after matching are presented in eTable 1, and the propensity score overlap is illustrated in eFigure 1.

Primary Outcome

Outcome variables and results of all regression models are summarized in Tables 2 and 3, respectively. The median 90-day mRS score was 3 (IQR 2–6) in both any previous APT

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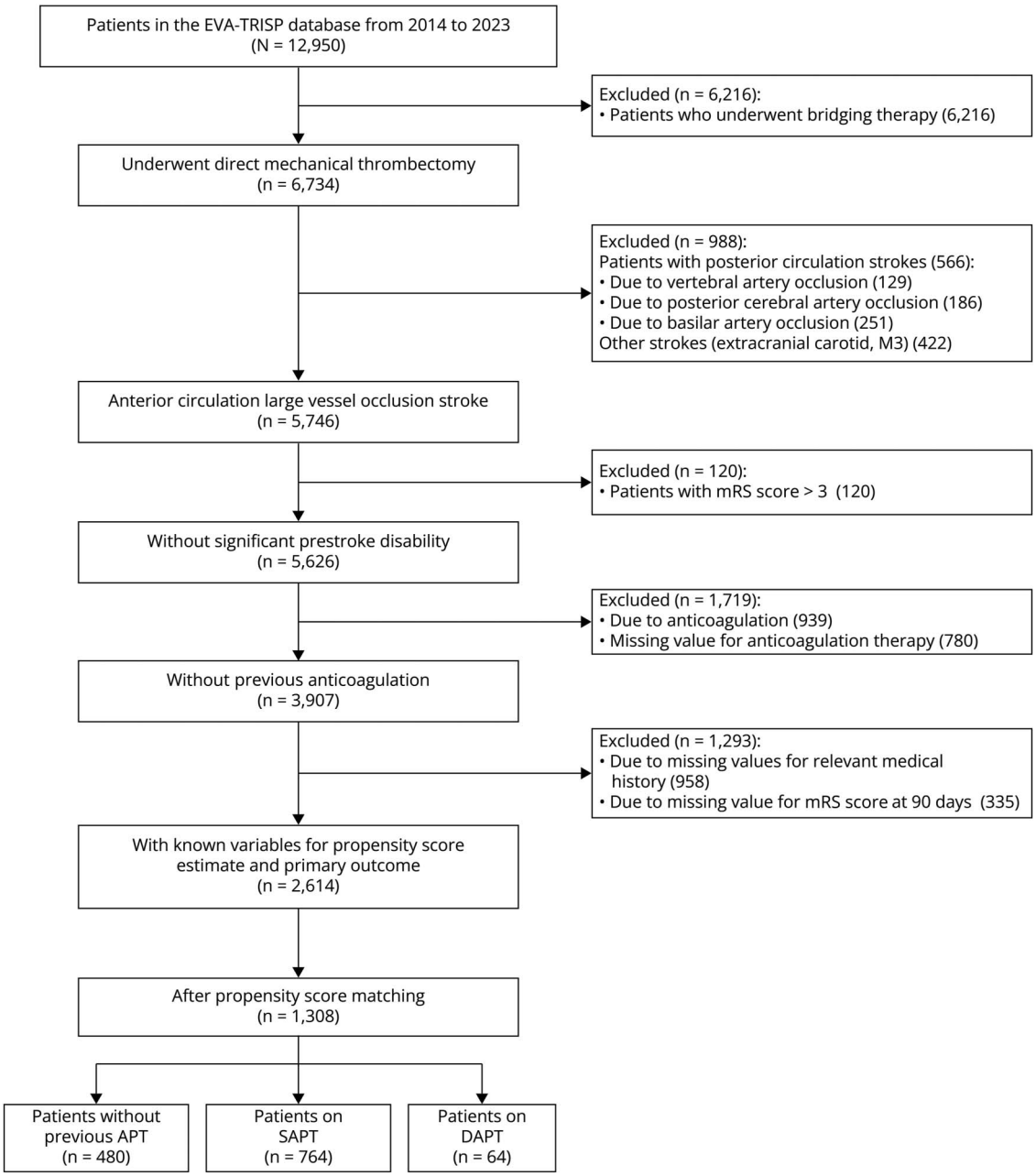
Figure 1 Study Cohort Flowchart



APT = antiplatelet therapy; DAPT = dual antiplatelet therapy; mRS = modified Rankin Scale; SAPT = single antiplatelet therapy.

and no previous APT groups. However, outcomes were skewed toward lower mRS scores in the any previous APT group (Figure 2). Indeed, in the multivariable ordinal analysis, any previous APT was associated with lower mRS scores (OR 1.30, CI 1.04–1.61, $p = 0.018$) when compared with no APT.

Secondary Outcomes

The percentage of patients who achieved 90-day independence was 41.8% in the any previous APT group and 33.8% in the no previous APT group. Any previous APT was associated with independence at 90 days (OR 1.62, CI 1.22–2.16, $p = 0.001$), when compared with no previous APT. The percentage of successful reperfusion was 75.9% in the any previous APT group and 72.7% in the no previous APT group, with no difference between the groups (OR 0.96, CI 0.69–1.35, $p = 0.82$). The percentage of patients with sICH was 5.2% in the any previous APT group and 6.7% in the no previous APT group (OR 1.06, CI 0.47–2.39, $p = 0.88$). Among patients aged 80 years or older, sICH rates were 5.08% in the any previous APT group and 4.06% in the no previous APT group. Mortality at 90 days after stroke was 25.4% in the any previous APT group and 27.3% in the no previous APT group (OR 0.89, CI 0.66–1.21, $p = 0.88$).

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Table 1 Baseline Characteristics After Propensity Score Matching

	No previous APT (N = 480)	Any previous APT (N = 828)
Demographics		
Age (y)	78 (69–84)	77 (69–84)
Sex (female)	221 (46)	425 (51.3)
Medical history		
Baseline mRS score	0 (0–1)	0 (0–1)
Atrial fibrillation	166 (34.6)	265 (32)
Diabetes	117 (24.4)	228 (27.5)
Hypertension	387 (80.6)	792 (84.8)
Hypercholesterolemia	258 (53.8)	536 (64.7)
Previous ischemic stroke	80 (16.7)	192 (23.2)
Prestroke statins	161 (33.5)	464 (56)
Prestroke antihypertensive	347 (72.3)	637 (76.9)
Current smoking	89 (18.6)	173 (20.9)
Coronary artery disease	56 (11.6)	324 (39.1)
Vitals and laboratory at admission		
Systolic BP (mm Hg)	153.6 ± 27.6	153.6 ± 29.7
Diastolic BP (mm Hg)	83.4 ± 17.4	80.6 ± 17.6
Platelets ($\times 10^9$/L)	239.1 ± 87.1	234 ± 89.5
Glucose (mmol/L)	7.6 ± 2.8	7.7 ± 2.8
Stroke and treatment characteristics		
NIHSS score at admission	13 (8–19)	13 (7–18)
ASPECTS	9 (8–10)	9 (7–10)
Wake-up stroke	132 (27.5)	231 (27.9)
Stroke onset witnessed	208 (43.3)	368 (44.4)
Onset to groin puncture (hours)	5.5 (3.0–11.3)	5.7 (3.2–10.2)
Early reocclusion	5 (1.1)	3 (0.4)
Etiologic classification		
Large artery atherosclerosis	92 (19.2)	204 (24.6)
Cardioembolism	183 (38.1)	334 (40.3)
Small artery occlusion	3 (0.6)	7 (0.8)
Other determined	81 (16.9)	93 (11.2)
Undetermined causes	121 (25.2)	190 (22.9)
Occlusion site		
ICA	119 (24.8)	211 (25.5)
MCA (M1)	262 (54.5)	411 (49.7)
MCA (M2)	126 (26.2)	249 (30.1)
ACA (A1)	5 (1.1)	9 (1.1)

Abbreviations: ACA = anterior cerebral artery; APT = antiplatelet therapy; ASPECTS = Alberta Stroke Program Early CT Score; BP = blood pressure; ICA = internal carotid artery; IQR = interquartile range; MCA = middle cerebral artery; mRS = modified Rankin Scale; NIHSS = NIH Stroke Scale.
Numbers are percentages for categorical variables, means ± SD for standard distributed variables, and median and IQR for ordinal variables.

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Sensitivity Analyses

When compared with no previous APT, any previous APT was associated with lower 90-day mRS scores even without correcting for center as a random effect (OR 1.30, CI 1.04–1.61, $p = 0.017$), and independently of the pretreatment probability of bleeding (OR 1.29, CI 1.04–1.60, $p = 0.023$), as expressed by the IVT exclusion criteria based on "imaging criteria associated with increased risk of hemorrhage" and "exclusion by clinician-assessed bleeding risk." In a post hoc sensitivity analysis, any previous APT was associated with lower 90-day mRS scores independently of coronary artery disease (OR 1.27, CI 1.61–1.01, $p = 0.046$) and previous statin therapy (OR 1.28, CI 1.59–1.02, $p = 0.032$). Previous statin therapy was not associated with lower 90-day mRS scores (OR 1.08, CI 1.34–0.87, $p = 0.491$).

Subgroup Analyses

Patients were classified based on previous antiplatelet regimen (SAPT [n = 764] or DAPT [n = 64]). The median 90-day mRS score was 3 (IQR 2–6) in the previous SAPT group and 2 (IQR 1–4) in the previous DAPT group. Previous SAPT was associated with a lower mRS score at 90 days compared with no previous APT (OR 1.28, CI 1.03–1.60, $p = 0.026$), whereas there was no association for previous DAPT compared with no previous APT (OR 1.55, CI 0.93–2.58, $p = 0.092$). The independence rate was 23.8% in the previous SAPT cohort and 31.2% in the previous DAPT cohort. Successful reperfusion on digital subtraction angiography was achieved in 75.5% and 80% of patients in previous SAPT and previous DAPT cohorts, respectively.

We also tested for effect modification based on occlusion site (proximal or ICA + M1 arteries, n = 833; distal M2 arteries, n = 329) and on time-to-groin (early <6 hours, n = 654; late treatment ≥ 6–24 hours, n = 565). There was no significant interaction between any previous APT and occlusion location (OR 1.39, CI 0.82–2.36, $p = 0.213$) or between previous APT use and the time-to-treatment group (OR 0.92, CI 0.60–1.43, $p = 0.743$).

Classification of Evidence

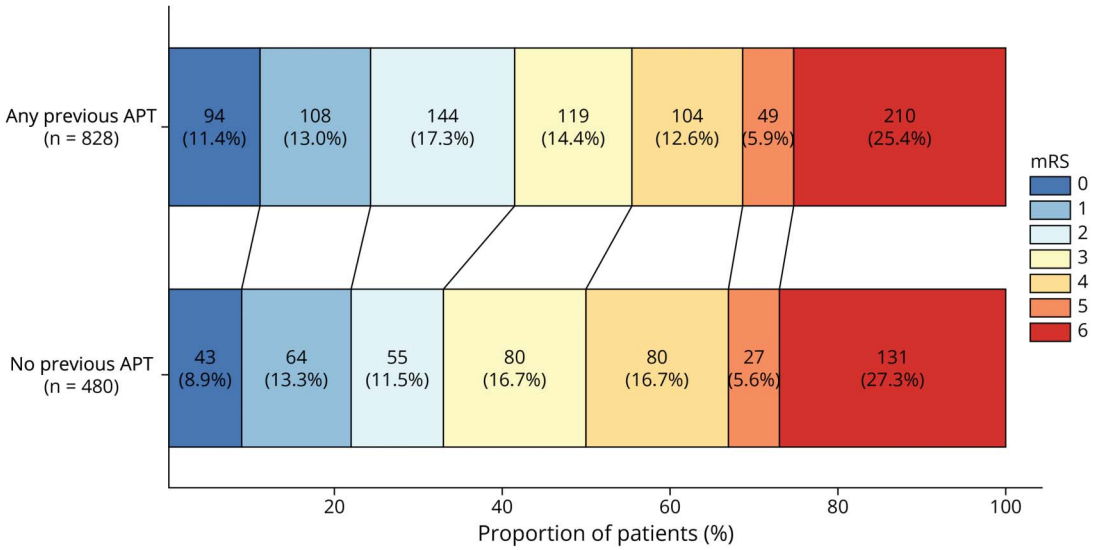
This study provides Class III evidence that, in patients with anterior circulation LVO stroke treated with direct EVT, previous APT was associated with better 90-day functional outcomes compared with no previous APT.

Discussion

Our study found that among patients diagnosed with ischemic stroke due to anterior circulation LVO who were treated with direct EVT, those who underwent previous APT had better functional outcomes on the shift analysis and significantly higher odds of 90-day independence, compared with those without previous APT. This result is significantly driven by the previous SAPT group.

The better outcome in patients with previous APT is not explained by a superior large vessel reperfusion rate. We observed a negligibly higher successful reperfusion rate in patients with previous SAPT (75.9%) vs those without previous APT regimen (72.7%): the difference is not statistically significant, a result in accordance with existing data.[7,8,13] Because the better outcome may not be explained by a large vessel reperfusion mechanism, we can, therefore, hypothesize that the results observed in our population could be related to a better microvascular reflow phenomenon.[20] The concept of microvascular

Figure 2 mRS Scores at 90 Days According to Treatment Groups



Stacked bar chart showing the distribution and percentage of each modified Rankin Scale (mRS) score at 90 days among patients treated with direct endovascular therapy (EVT) with previous antiplatelet therapy (APT) vs without previous APT. Direct EVT with any APT was associated with a shift toward lower mRS scores at 90 days (odds ratio 1.30, CI 1.04–1.61, $p = 0.018$).

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Table 2 Outcome Measures After Propensity Score Matching

	No previous APT (N = 480)	Any previous APT (N = 828)
Efficacy outcomes		
90-d mRS score	3 (2–6)	3 (2–6)
Secondary efficacy outcomes		
90-d independence (mRS score 0–2)	164 (33.8)	346 (41.8)
Successful reperfusion on DSA (mTICI score 2b–3)	353 (72.7)	628 (75.9)
Secondary safety outcomes		
In-hospital sICH (ECASS2)	33 (6.7)	43 (5.2)
90-d mortality	133 (27.3)	210 (25.4)

Abbreviations: APT = antiplatelet therapy; DSA = digital subtraction angiography; ECASS2 = European Co-operative Acute Stroke Study-II; IQR = interquartile range; mRS = modified Rankin Scale; mTICI = modified Thrombolysis in Cerebral Infarction; sICH = symptomatic intracranial hemorrhage.
Numbers are percentages for categorical variables and median and IQR for ordinal variables.

no-reflow, or impaired tissue reperfusion despite successful large vessel recanalization, may offer a theoretical explanation for the observed benefits in the previous APT group in our study. In theory, an antiplatelet agent could help mitigate distal microthrombus formation and platelet aggregation, both of which are implicated in no-reflow phenomena. Nevertheless, this hypothetical explanation remains unproven and not supported by our data. Recent randomized clinical trials addressing microvascular no-reflow in thrombectomy investigated the effect of fibrinolytic agents as an add-on to successful reperfusion,[21,22] but the relationship between microvascular reperfusion and APT has not been addressed in these trials. This aspect should be studied in future dedicated trials.

It is important to note that previous APT was safe in patients with direct EVT, and we found no association between previous APT and sICH. This result aligns with previous retrospective studies[1-3,5-10] and contrasts with the main finding of the MR CLEAN MED trial.[12] Various factors might explain this difference: First, the MR CLEAN MED trial included a relevant number of patients on bridging with IVT (75% of cases) and did not exclude patients previously on anticoagulants or those with previous APT. Second, patients were randomized into both study drugs with the potential of receiving them both simultaneously, which results in additional confounding.

We also performed a subgroup analysis based on location and treatment timing: proximal (ICA and M1) vs distal (M2 segment) occlusions; early (<6 hours) vs late (≥6–24 hours) time-to-groin. We observed that there was no statistical difference based on occlusion location (proximal vs distal) or due to treatment timing (early vs late time-to-groin) between patients with no previous APT or any previous APT.

Patients on previous DAPT had the lowest median 90-day mRS score, the highest rate of independence, and the highest rate

Table 3 Primary and Secondary Outcome Analysis

	Any previous APT (vs no previous APT) Adjusted OR (95% CI)
Primary outcome	
Lower mRS score	1.30 (1.04–1.61)[a]
Secondary efficacy outcomes	
90-d independence (mRS score 0–2)	1.62 (1.22–2.16)[a]
Successful reperfusion on DSA (mTICI score 2b–3)	0.96 (0.69–1.35)
Secondary safety outcomes	
In-hospital sICH (ECASS 2)	1.06 (0.47–2.39)
90-d mortality	0.89 (0.66–1.21)

Abbreviations: APT = antiplatelet therapy; DSA = digital subtraction angiography; ECASS2 = European Co-operative Acute Stroke Study-II; mRS = modified Rankin Scale; mTICI = modified Thrombolysis in Cerebral Infarction; OR = odds ratio; sICH = symptomatic intracranial hemorrhage.
[a] Significant differences compared with the no previous APT group.

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of reperfusion compared with previous SAPT. However, the association between previous DAPT and lower 90-day mRS score was not statistically significant, and this was likely due to the much smaller sample size of the previous DAPT group compared with the previous SAPT group. Overall, the results of the DAPT subgroup should be interpreted with caution because of its small sample size (n = 64). The role of DAPT in the acute treatment of stroke, as an alternative to IVT, and its potential future use as an adjunctive therapy for direct EVT remains speculative. This question warrants further exploration in dedicated future trials.

Our study has strengths: First, in contrast to previous studies[1-12] which included patients treated with bridging (IVT + EVT), our dedicated study compares the outcomes of patients treated with direct EVT only. By excluding patients treated with IVT and patients with previous anticoagulation, we avoided any potential confounding effect on both efficacy and safety outcomes (i.e., risk of bleeding). Second, we analyzed data from a large international multicentric database, with detailed data collection and inclusiveness of real-world patient populations. Third, we addressed confounders with propensity score matching and multilevel regression analysis. Finally, we conducted a sensitivity analysis and adjusted for the IVT exclusion criteria, which resulted in stable associations independent of the pretreatment probability of bleeding.

Despite its strengths, our study also has limitations: First, we analyzed retrospective data, excluding many patients because of missing values. Second, selection bias may still affect the analysis: despite propensity score matching, we found some differences in baseline characteristics (age and atrial fibrillation), as well as differences in stroke etiology and onset to groin puncture. Third, information on adherence to previous APT therapy was not available; therefore, given the retrospective nature of the study, a patient may have been incorrectly categorized with respect to their previous APT status, and this could have influenced the observed associations.

In conclusion, this study provides observational evidence that previous APT is associated with better functional outcomes after direct EVT used to treat acute ischemic stroke patients with LVO of the anterior circulation without previous use of anticoagulants. This finding is independent of occlusion location, treatment timing (expressed by the time-to-groin), and the pretreatment probability of bleeding. Regarding efficacy, successful reperfusion was not influenced by previous APT regimen; although unproven by our study, better microvascular reflow could explain a possible mechanistic link for the better functional outcome. Regarding safety, sICH rate and mortality were not influenced by previous APT regimen. We deem it important to test this hypotheses on further observational data sets and/or in randomized controlled trials.

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Author Contributions

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